

Toelichting constructie van PDA naar CFG

Wat betekenen die nieuwe regels?

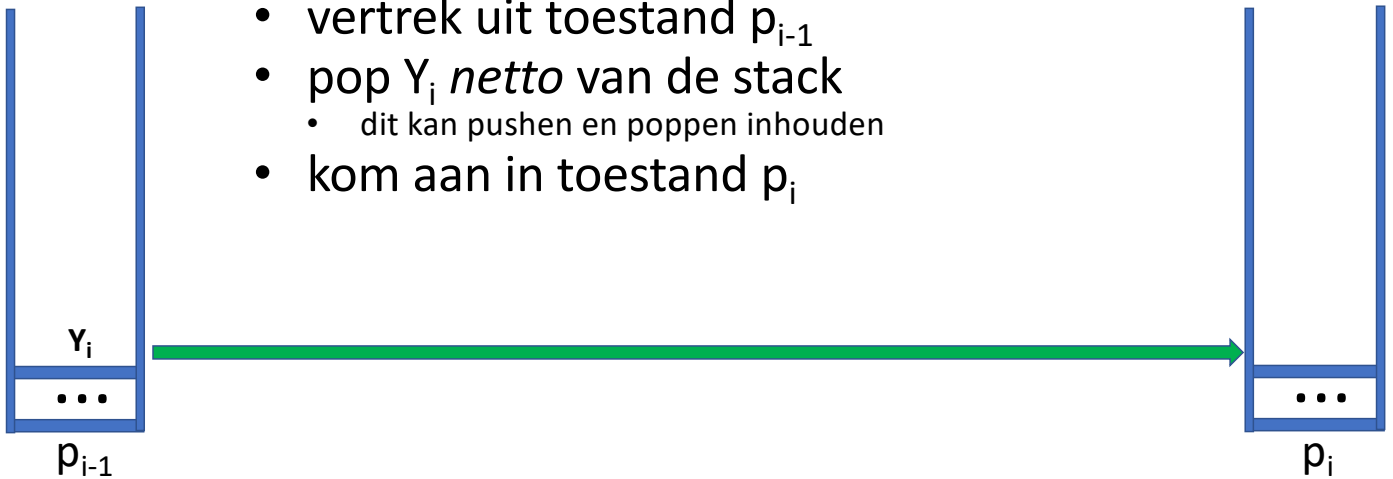
Formally, let $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0)$ be a PDA.
Define $G = (V, \Sigma, R, S)$, where

$$\begin{aligned} V &= \{[pXq] : \{p, q\} \subseteq Q, X \in \Gamma\} \cup \{S\} \\ R &= \{S \rightarrow [q_0Z_0p] : p \in Q\} \cup \\ &\quad \{ \underline{[qXr_k] \rightarrow a[rY_1r_1][r_1Y_2r_2] \cdots [r_{k-1}Y_kr_k]} : \\ &\quad \quad a \in \Sigma \cup \{\epsilon\}, \\ &\quad \quad \{r_1, \dots, r_k\} \subseteq Q, \\ &\quad \quad (r, Y_1Y_2 \cdots Y_k) \in \delta(q, a, X) \} \\ &\quad \cup \{ \underline{[qXr] \rightarrow a} : a \in \Sigma \cup \{\epsilon\}, \\ &\quad \quad (r, \epsilon) \in \delta(q, a, X) \} \end{aligned}$$

$$[p_{i-1} Y_i p_i]$$

Deze variabele komt overeen met

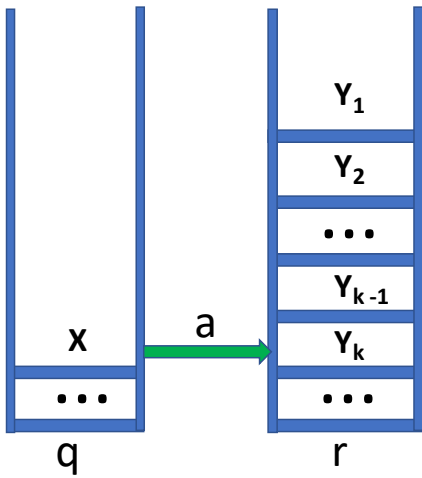
- vertrek uit toestand p_{i-1}
- pop Y_i *netto* van de stack
 - dit kan pushen en poppen inhouden
- kom aan in toestand p_i



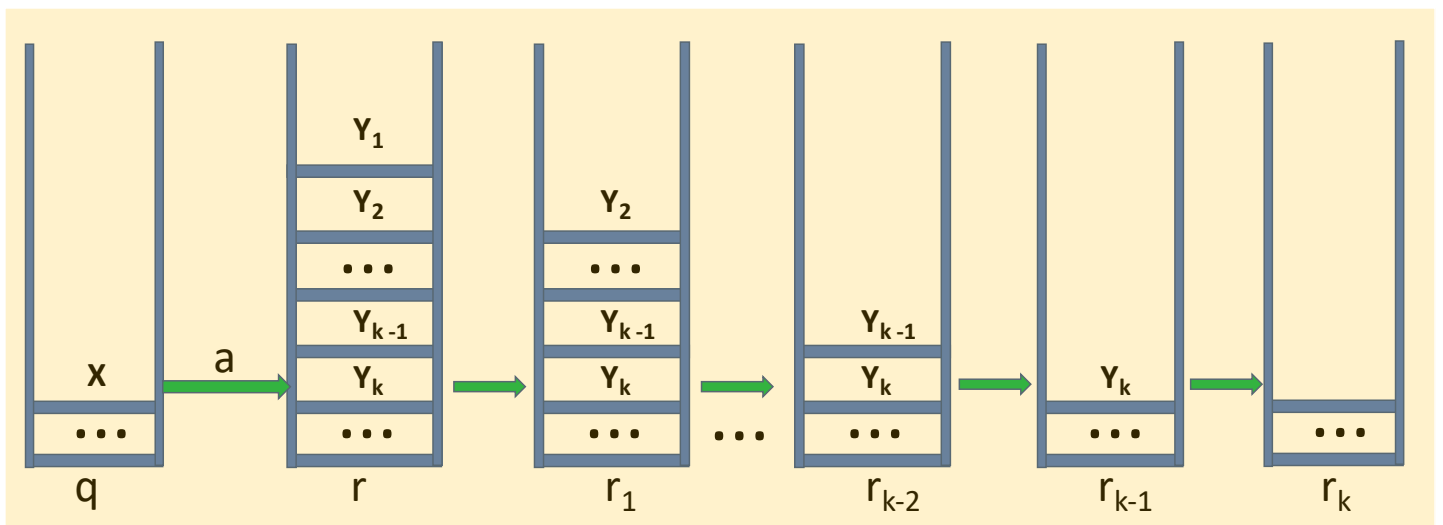
$$[q X r_k]$$

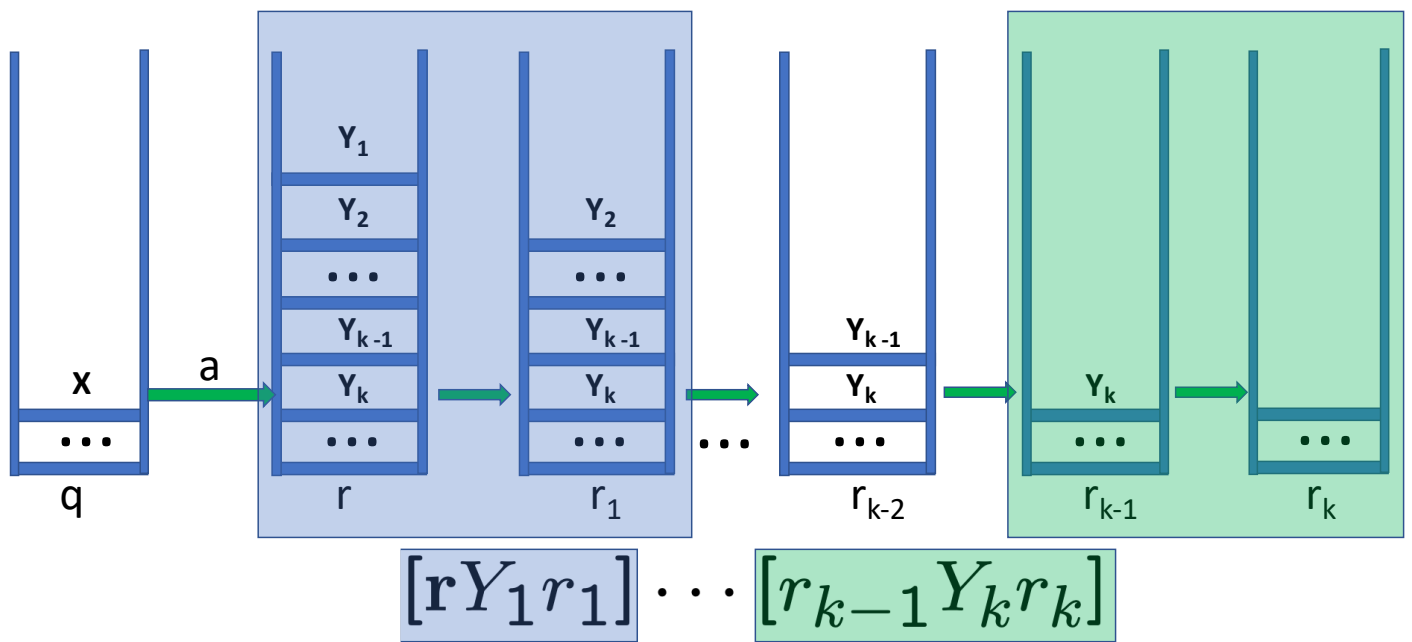


$$(r, Y_1 Y_2 \cdots Y_k) \in \delta(q, a, X)$$

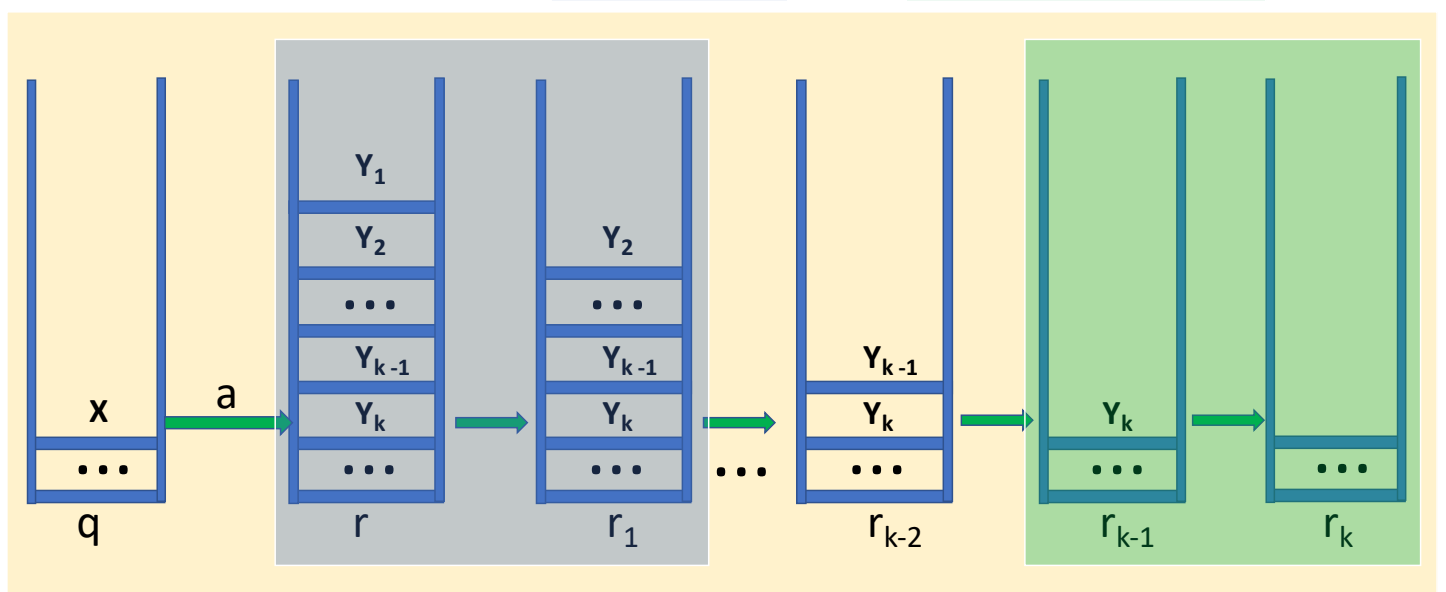


$$[qXr_k]$$

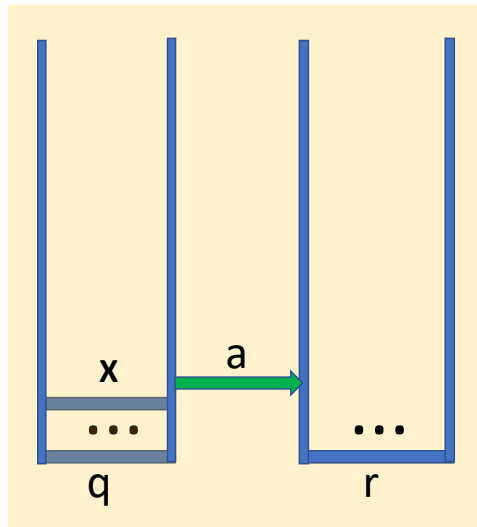




$$[qXr_k] \rightarrow a[rY_1r_1] \cdots [r_{k-1}Y_kr_k]$$



$$(\mathbf{r}, \epsilon) \in \delta(\mathbf{q}, a, X)$$



$$[qXr] \rightarrow a$$